

1 Easy

Problem 1 (Source: ???)

Let $P(x)$ be a cubic polynomial with rational coefficients such that

$$P(\sqrt{2}) + \sqrt{2}, \quad \sqrt{3}P(\sqrt{3}) + \sqrt{3}, \quad P(\sqrt{5}) - \sqrt{5}, \quad \sqrt{7}P(\sqrt{7}) - \sqrt{7}$$

are all rational. Compute $P(1)$.

2 Medium-Easy

Problem 2 (Source: ???)

Consider the polynomial $P(x) = x^{400} + 12x + 1$ with roots $r_1, r_2, \dots, r_{400}$. Let $Q(x)$ be the monic polynomial whose roots are exactly $r_i^4 + 2$ for $i = 1, 2, \dots, 400$. Compute $Q(1)$.

3 Medium

Problem 3 (Source: ???)

Let a, b , and c be the roots of the polynomial $x^3 + 12x^2 + 20x + 24$. There exists a monic cubic polynomial $P(x)$ such that

$$P(a) = \frac{b+c}{a}, \quad P(b) = \frac{c+a}{b}, \quad \text{and} \quad P(c) = \frac{a+b}{c}.$$

Compute $P(0)$.

4 Medium

Problem 4 (Source: ???)

Let F_k denote the k th Fibonacci number defined by $F_1 = F_2 = 1$ and $F_k = F_{k-1} + F_{k-2}$ for $k \geq 2$. Suppose that $P(x)$ is the unique polynomial with degree 99 satisfying $P(k) = F_k$ for $k = 1, 2, 3, \dots, 100$. Find $P(101)$.

5 Hard-Medium

Problem 5 (Source: ???)

Let $f(x) = x^2 - 3x + 1$ such that $\alpha_1, \alpha_2, \alpha_3$, and α_4 are the roots to $f(f(x)) = x$. Find

$$\lfloor 10\alpha_1 \rfloor + \lfloor 10\alpha_2 \rfloor + \lfloor 10\alpha_3 \rfloor + \lfloor 10\alpha_4 \rfloor.$$

6 Hard

Problem 6 (Source: ???)

Let a , b , and c be roots of the polynomial $x^3 - x^2 - 2x + 1$. Let S be the set of triples of non-negative integers (i, j, k) with $i + j + k = 9$ and $ijk = 0$. Compute

$$\sum_{(i,j,k) \in S} a^i b^j c^k.$$